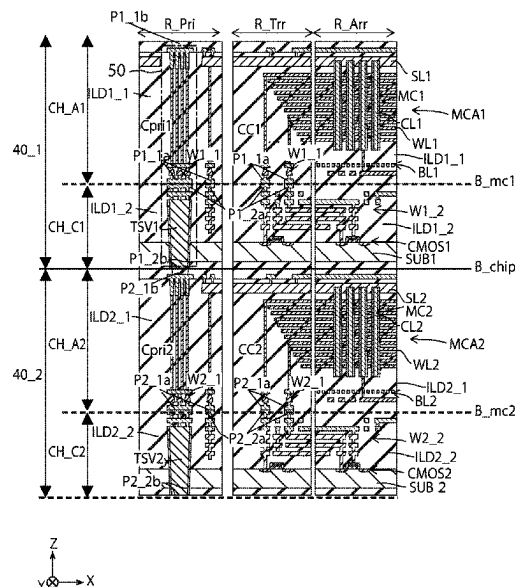
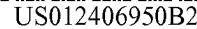




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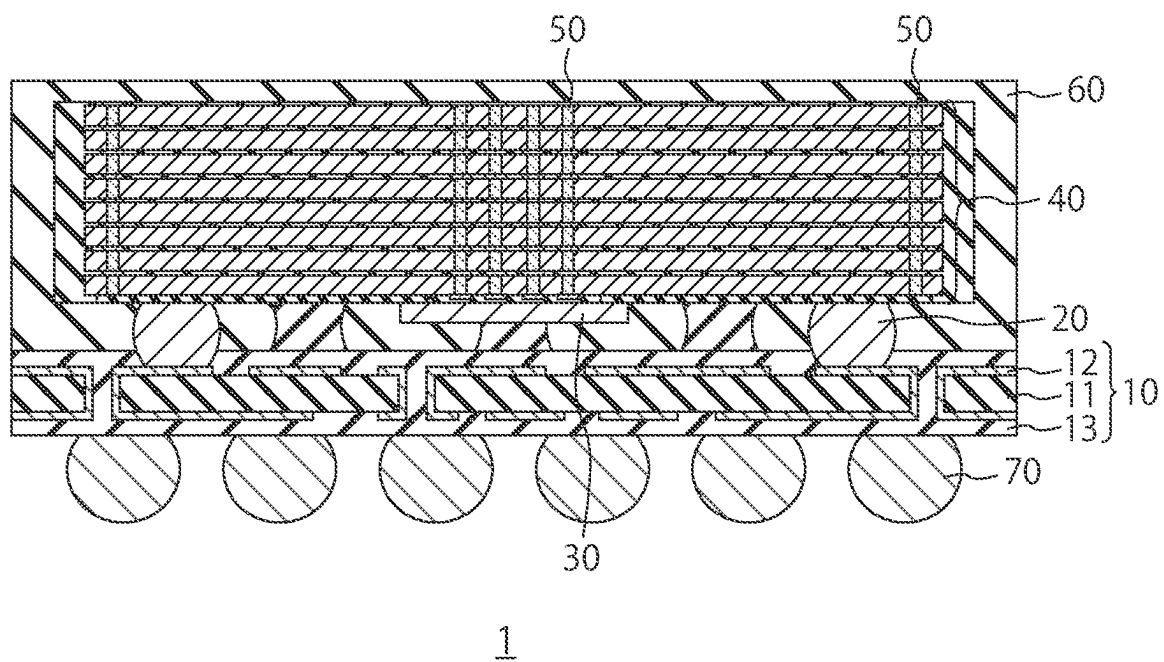


FIG. 1

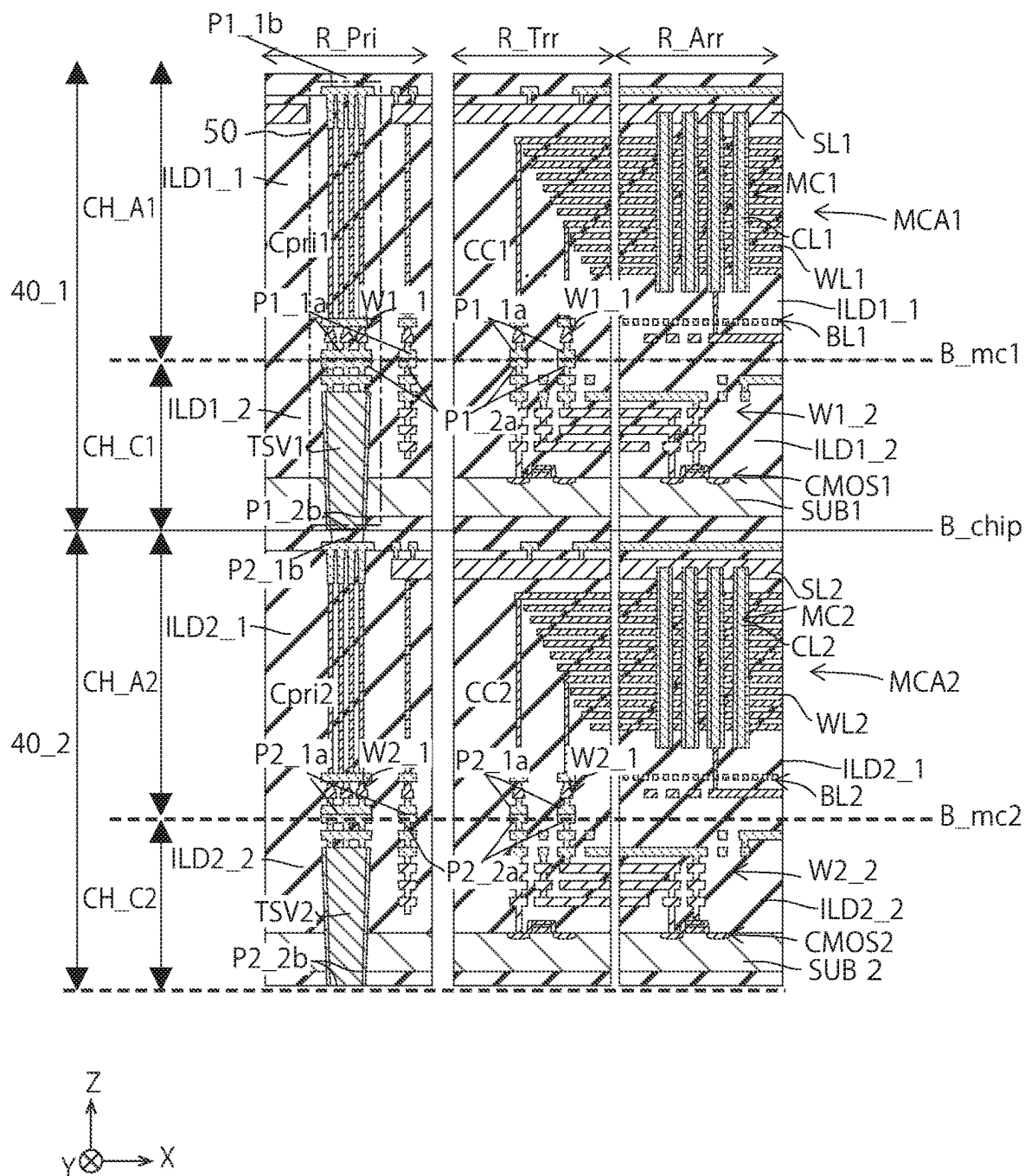


FIG. 2

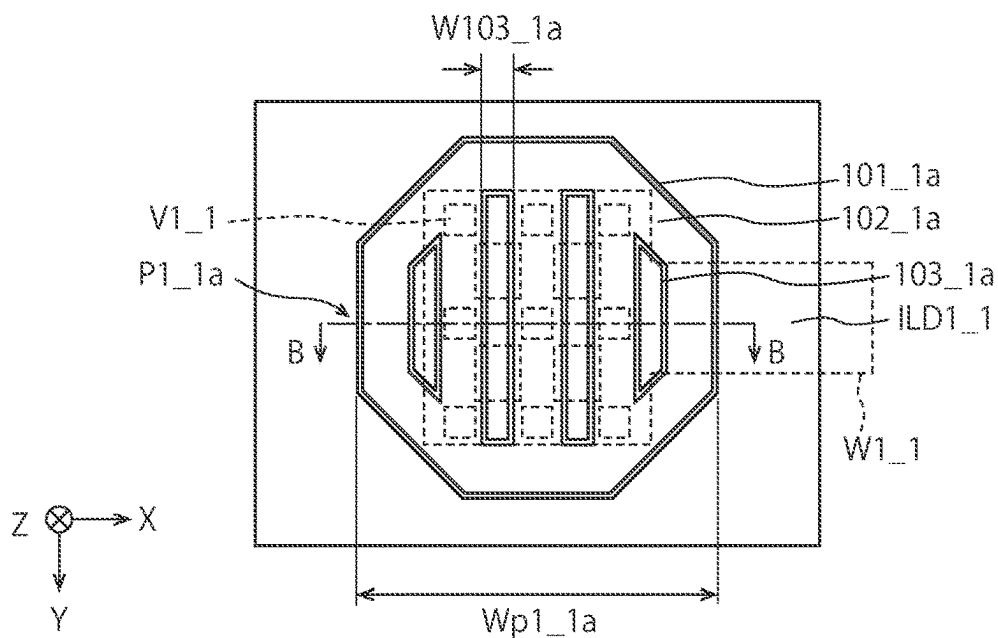


FIG. 3A

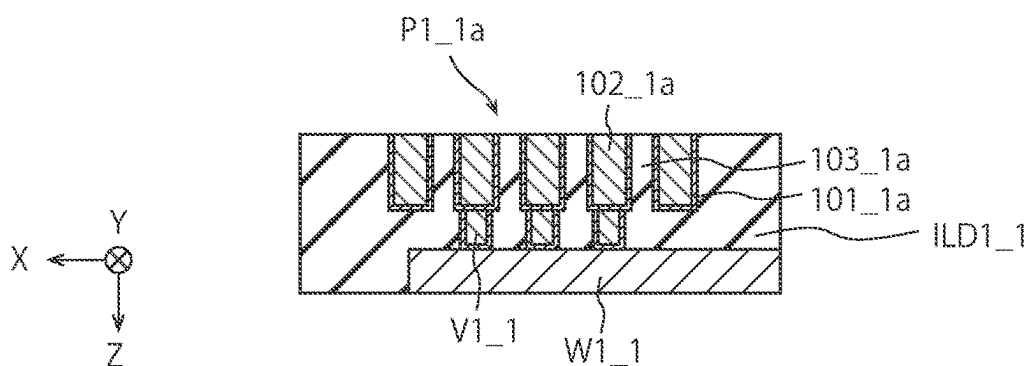


FIG. 3B

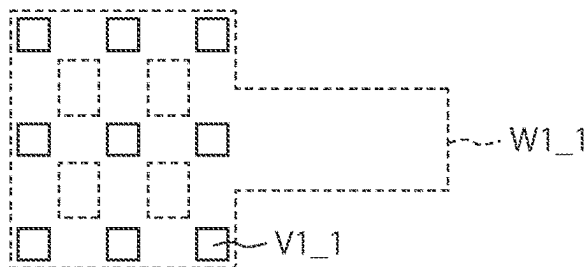


FIG. 3C

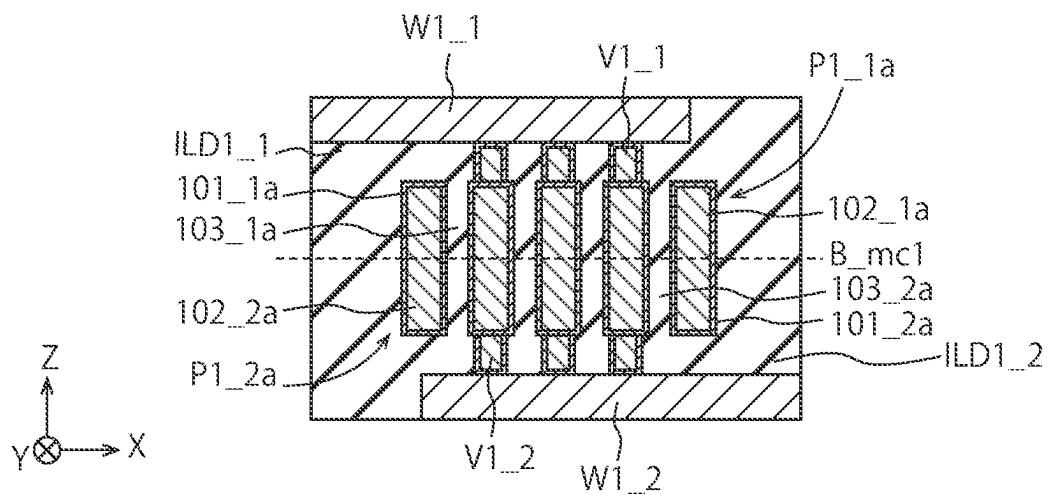


FIG. 4

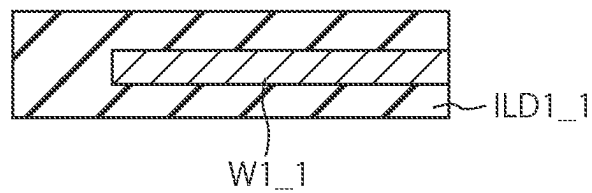


FIG. 5

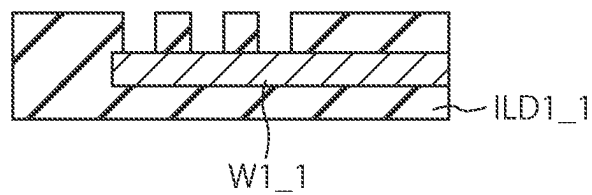


FIG. 6

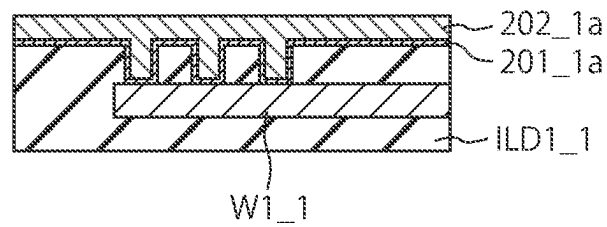


FIG. 7

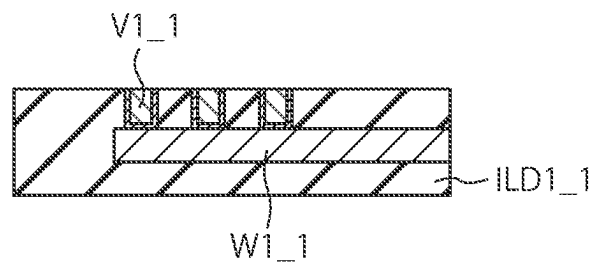


FIG. 8

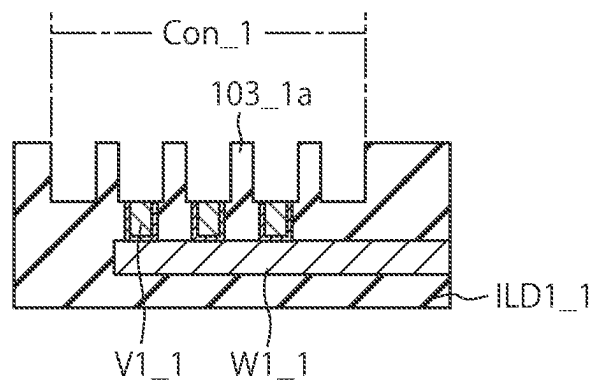


FIG. 9

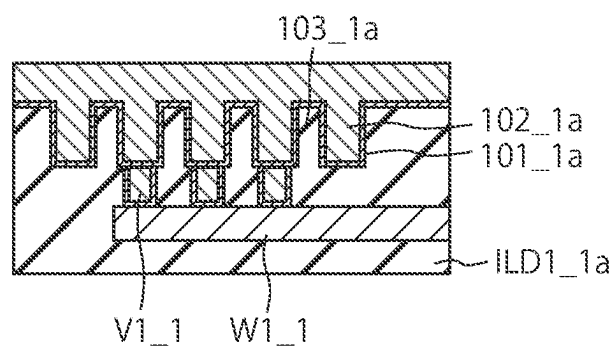


FIG. 10

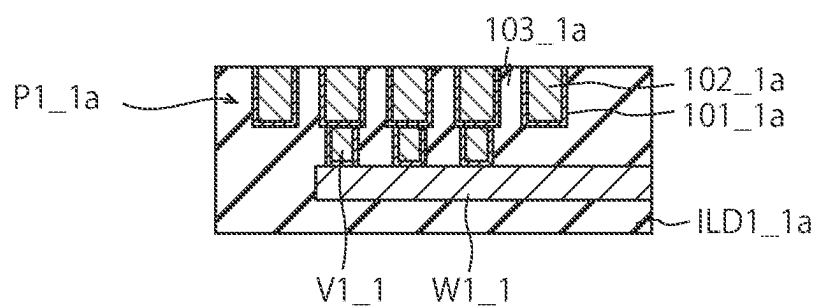


FIG. 11

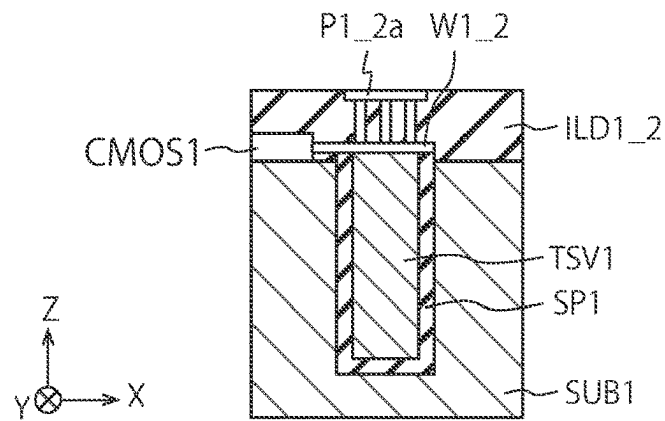


FIG. 12

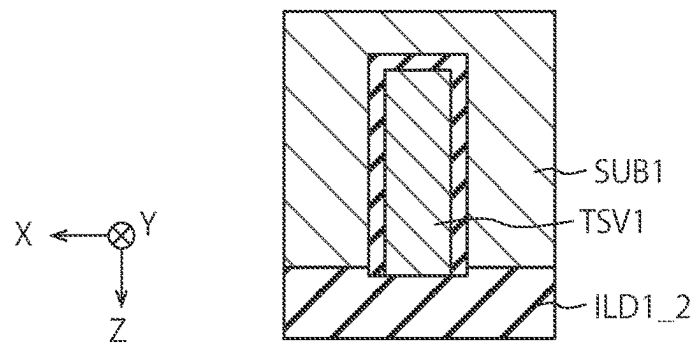


FIG. 13

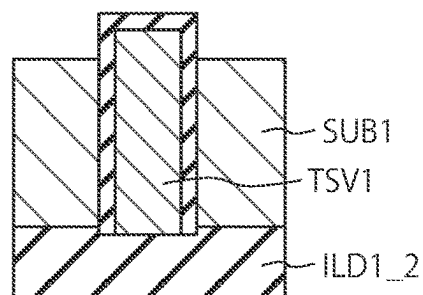


FIG. 14

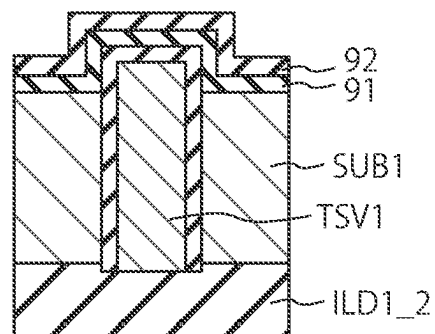


FIG. 15

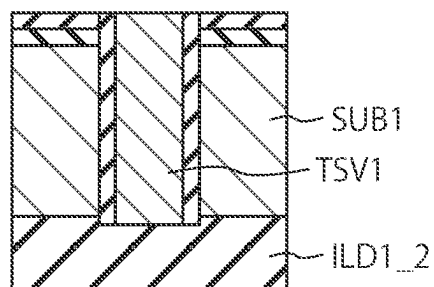


FIG. 16

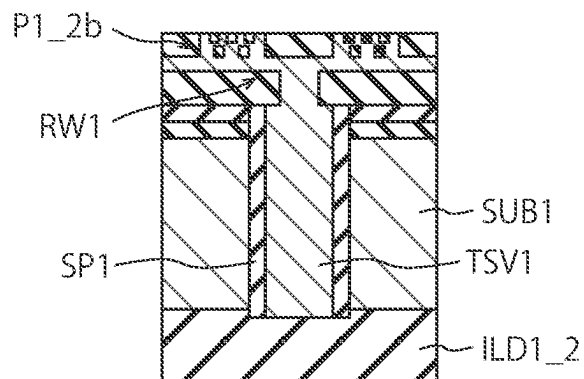


FIG. 17

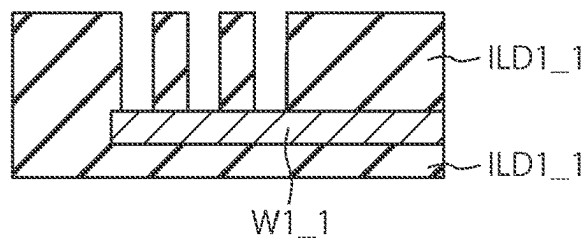


FIG. 18A

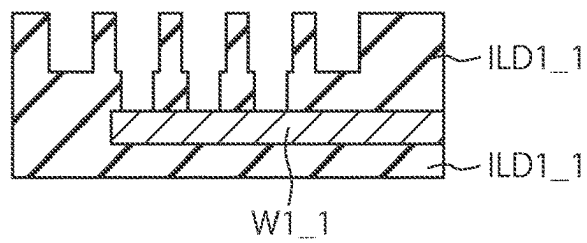


FIG. 18B

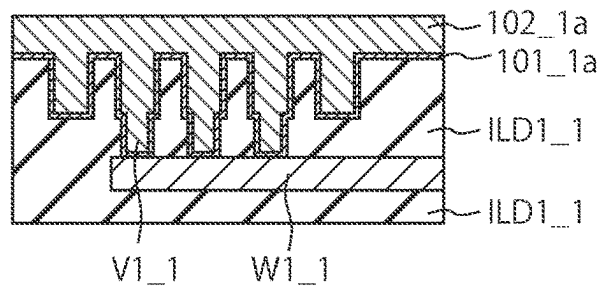


FIG. 18C

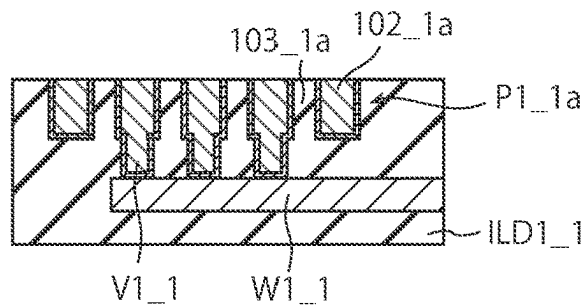


FIG. 18D

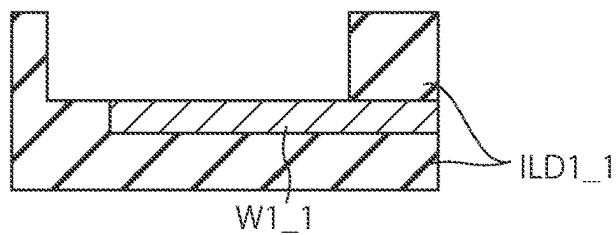


FIG. 19A

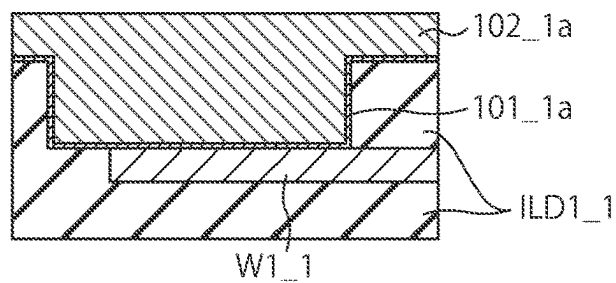


FIG. 19B

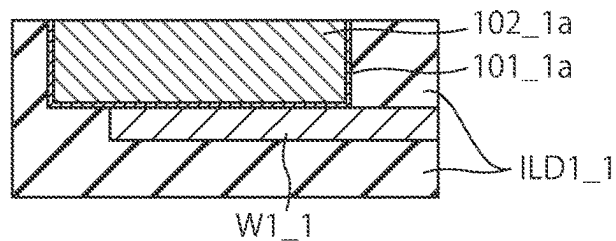


FIG. 19C

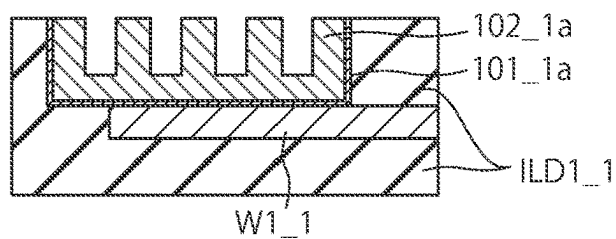


FIG. 19D

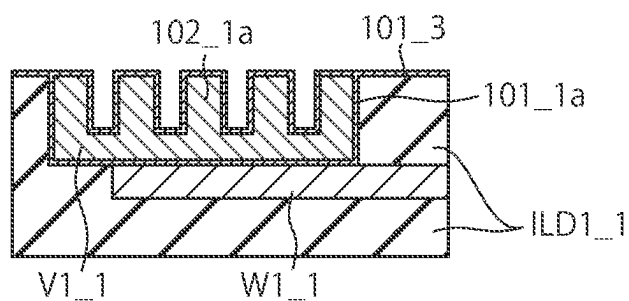


FIG. 19E

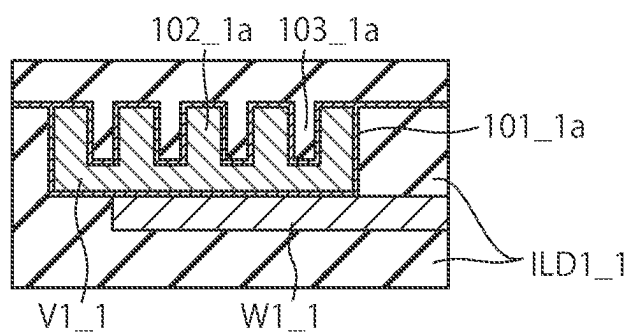


FIG. 19F

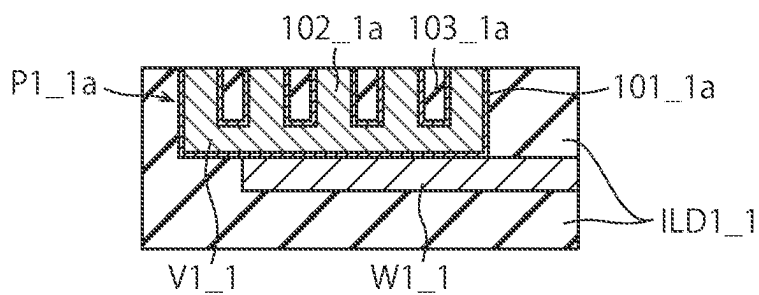


FIG. 19G

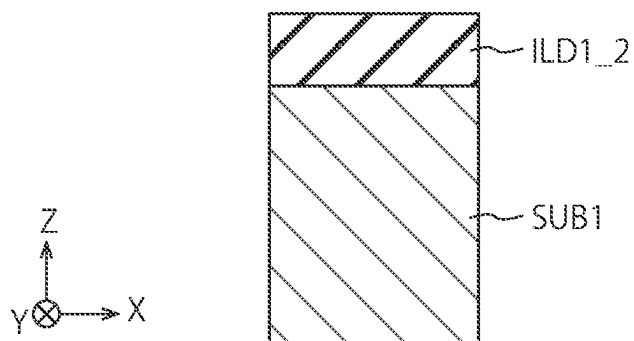


FIG. 20A

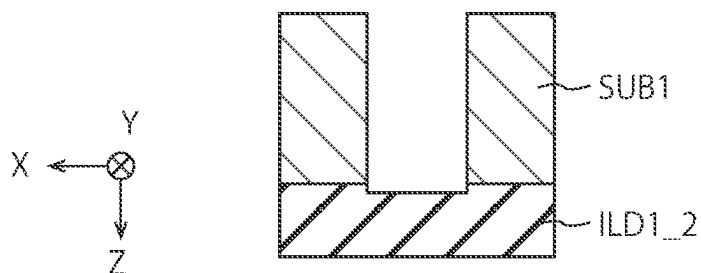


FIG. 20B

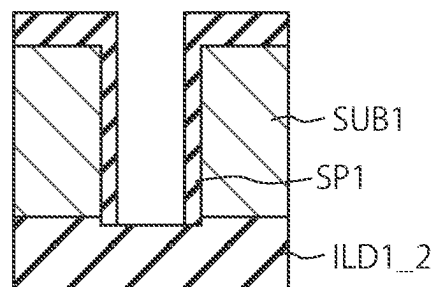


FIG. 20C

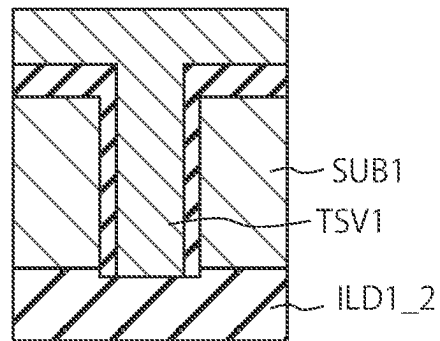


FIG. 20D

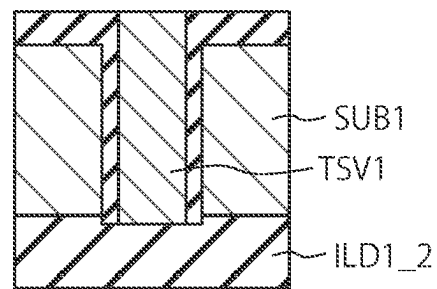


FIG. 20E

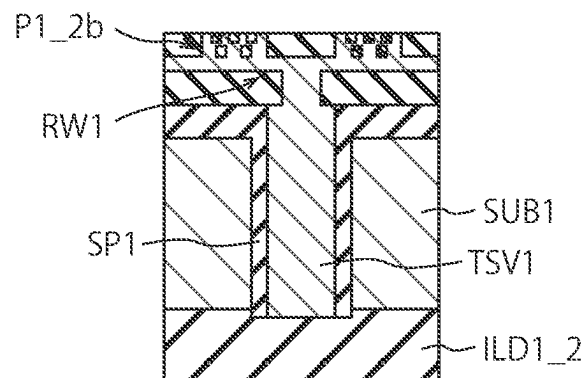


FIG. 20F

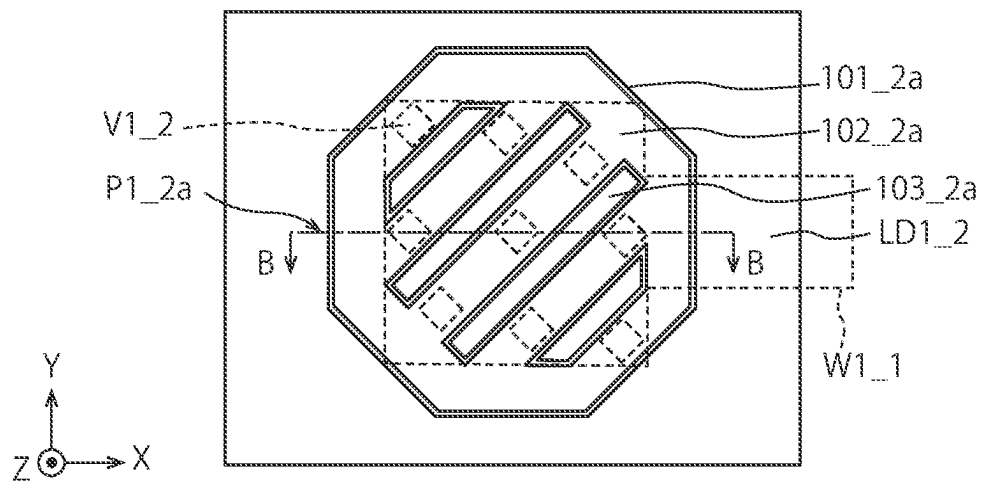


FIG. 21

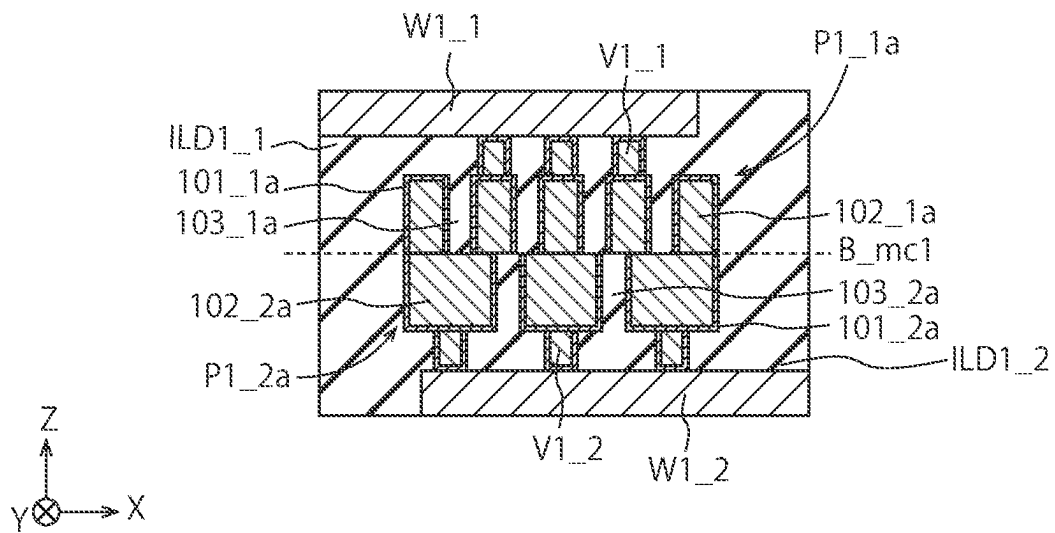


FIG. 22

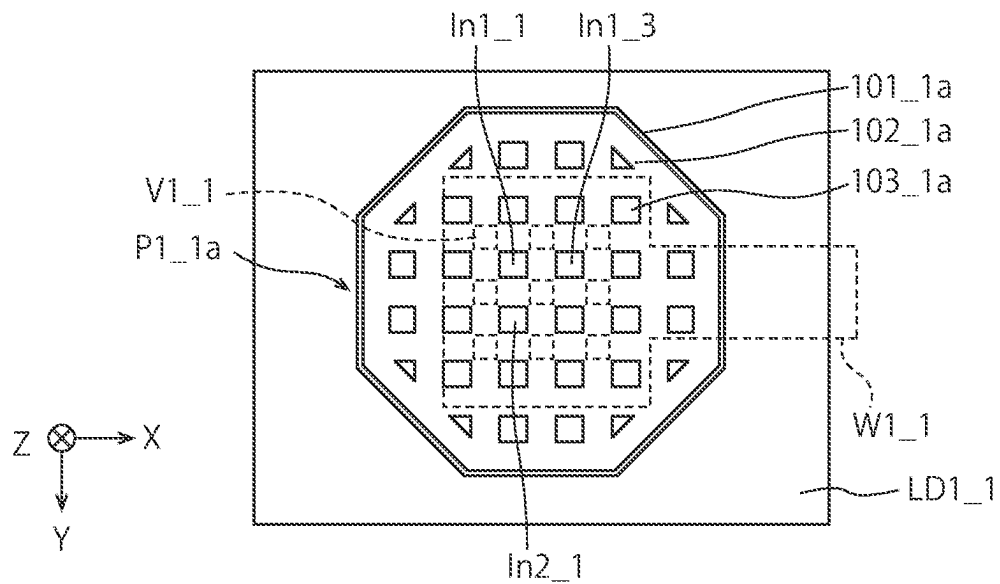


FIG. 23

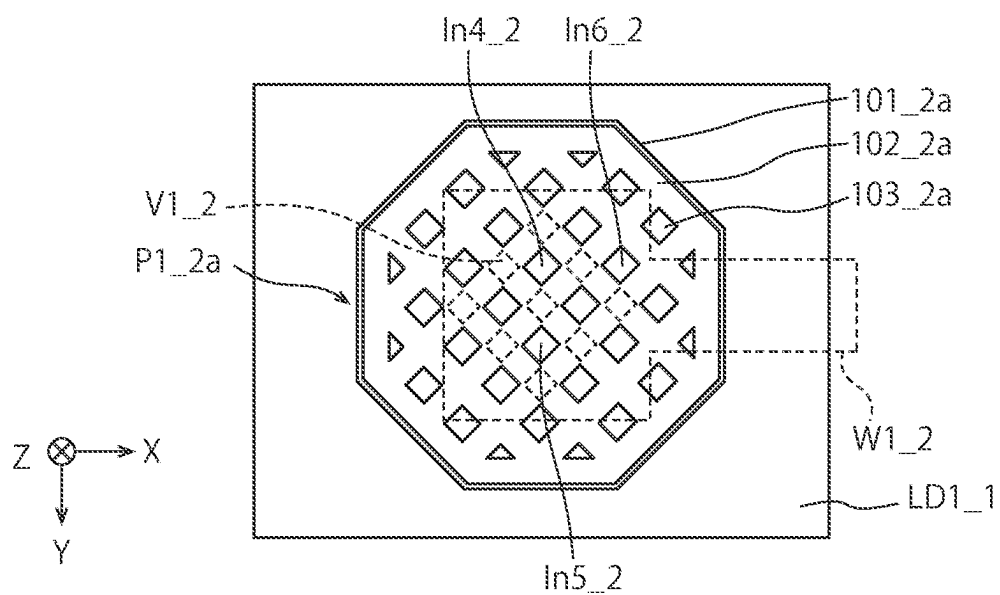


FIG. 24

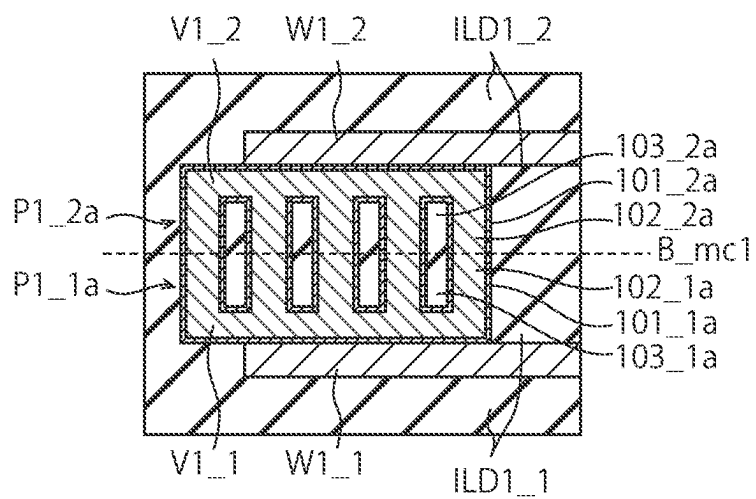


FIG. 25

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SEMICONDUCTOR DEVICE**CROSS REFERENCE TO RELATED APPLICATIONS**

This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2021-086411, filed on May 21, 2021, the entire contents of which are incorporated herein by reference.

FIELD

The embodiments of the present invention relate to a semiconductor device.

BACKGROUND

In recent years, a technology of sticking a plurality of semiconductor chips to electrically bond pads to each other is developed. Meanwhile, in a polishing method such as a CMP (Chemical Mechanical Polishing) method, dishing (a recess) sometimes occurs due to a difference in the quality of materials to be polished. If the pads on the bonding surface are recessed due to dishing, the contact resistance between the pads may be increased or an open circuit failure between the pads may occur when a plurality of semiconductor chips are stuck to each other.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a sectional view illustrating a configuration example of a semiconductor package according to a first embodiment;

FIG. 2 is a sectional view illustrating a configuration example of a portion of the semiconductor package according to the first embodiment;

FIG. 3A is a plan view illustrating a configuration example of pads;

FIG. 3B is a sectional view illustrating a configuration example of the pads;

FIG. 3C is a plan view illustrating one example of a configuration of a wiring layer;

FIG. 4 is a sectional view illustrating a configuration example of a portion of a bonding surface;

FIG. 5 is a sectional view illustrating one example of a manufacturing method of the pads according to the first embodiment;

FIG. 6 is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. 5;

FIG. 7 is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. 6;

FIG. 8 is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. 7;

FIG. 9 is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. 8;

FIG. 10 is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. 9;

FIG. 11 is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. 10;

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FIG. 12 is a sectional view illustrating one example of a formation process of a region of through silicon vias of a circuit chip;

FIG. 13 is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. 12;

FIG. 14 is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. 13;

FIG. 15 is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. 14;

FIG. 16 is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. 15;

FIG. 17 is a sectional view illustrating one example of the manufacturing method of the pads in continuation from FIG. 16;

FIG. 18A is a sectional view illustrating another example of the manufacturing method of the pads;

FIG. 18B is a sectional view illustrating the manufacturing method of the pads in continuation from FIG. 18A;

FIG. 18C is a sectional view illustrating the manufacturing method of the pads in continuation from FIG. 18B;

FIG. 18D is a sectional view illustrating the manufacturing method of the pads in continuation from FIG. 18C;

FIG. 19A is a sectional view illustrating still another example of the manufacturing method of the pads;

FIG. 19B is a sectional view illustrating the manufacturing method of the pads in continuation from FIG. 19A;

FIG. 19C is a sectional view illustrating the manufacturing method of the pads in continuation from FIG. 19B;

FIG. 19D is a sectional view illustrating the manufacturing method of the pads in continuation from FIG. 19C;

FIG. 19E is a sectional view illustrating the manufacturing method of the pads in continuation from FIG. 19D;

FIG. 19F is a sectional view illustrating the manufacturing method of the pads in continuation from FIG. 19E;

FIG. 19G is a sectional view illustrating the manufacturing method of the pads in continuation from FIG. 19F;

FIG. 20A is a sectional view illustrating another example of the formation process of the region of the through silicon vias of the circuit chip;

FIG. 20B is a sectional view illustrating another example of the formation process of the region of the through silicon vias in continuation from FIG. 20A;

FIG. 20C is a sectional view illustrating another example of the formation process of the region of the through silicon vias in continuation from FIG. 20B;

FIG. 20D is a sectional view illustrating another example of the formation process of the region of the through silicon vias in continuation from FIG. 20C;

FIG. 20E is a sectional view illustrating another example of the formation process of the region of the through silicon vias in continuation from FIG. 20D;

FIG. 20F is a sectional view illustrating another example of the formation process of the region of the through silicon vias in continuation from FIG. 20E;

FIG. 21 is a plan view illustrating a configuration example of pads according to a second embodiment;

FIG. 22 is a sectional view illustrating a configuration example of a region of a bonding surface according to the second embodiment;

FIG. 23 is a plan view illustrating a configuration example of pads according to a third embodiment;

FIG. 24 is a plan view illustrating a configuration example of pads according to a fourth embodiment; and

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FIG. 25 is a sectional view illustrating a configuration example of a region of a bonding surface according to a fifth embodiment.

DETAILED DESCRIPTION

Embodiments will now be explained with reference to the accompanying drawings. The present invention is not limited to the embodiments. In the embodiments, the term “upper direction” or “lower direction” of a semiconductor chip occasionally differs from an upper direction or a lower direction based on a gravitational acceleration direction. In the present specification and the drawings, elements identical to those described in the foregoing drawings are denoted by like reference characters and detailed explanations thereof are omitted as appropriate.

A semiconductor device according to the present embodiment includes a substrate having a first semiconductor circuit provided thereon. First pads are located on the substrate. A first insulating layer is located on an outer side of each of the first pads. Second pads are respectively bonded to the first pads. A second insulating layer is located on an outer side of each of the second pads and is bonded to the first insulating layer. The first pads each include a first conductive material, and a first insulating material located on an inner side of the first conductive material on a bonding surface of the first pads and the second pads.

First Embodiment

FIG. 1 is a sectional view illustrating a configuration example of a semiconductor package 1 according to a first embodiment. The semiconductor package 1 of the present embodiment is an example of a package of semiconductor memories. However, the present embodiment can also be applied to other semiconductor devices.

The semiconductor package 1 includes a wiring substrate 10, metallic bumps 20, solder balls 70, a controller chip 30, a memory chip stacked body 40 including a plurality of stacked memory chips, electrodes 50 provided to penetrate through the memory chips, and a sealing resin 60.

The wiring substrate 10 includes an insulator 11, a wiring layer 12, and a solder resist layer 13. For example, an insulating material such as glass epoxy resin is used as the insulator 11. The wiring layer 12 is a conductor provided on the front and back surfaces of the insulator 11. For example, a low-resistance metallic material such as copper is used as the wiring layer 12. The solder resist layer 13 is provided on the wiring layer 12.

The metallic bumps 20 are provided on the front surface side of the wiring substrate 10 and are electrically connected to portions of the wiring layer 12, respectively. The solder balls 70 are provided on the back surface side of the wiring substrate 10 and are electrically connected to portions of the wiring layer 12, respectively.

The controller chip 30 is provided above the front surface of the wiring substrate 10. The controller chip 30 is provided to control the memory chips.

The memory chip stacked body 40 is stacked on the controller chip 30. The memory chips are, for example, semiconductor chips on which NAND memory cells are mounted. The memory chips and the controller chip 30 are electrically connected via the electrodes 50. The electrodes 50 transmit supply power, a ground voltage, a control signal, data, or the like. For example, a conductive material such as tungsten, nickel, copper, gold, aluminum, or polysilicon is used as the electrodes 50.

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The sealing resin 60 is provided on the front surface of the wiring substrate 10 and seals the controller chip 30 and the memory chip stacked body 40.

FIG. 2 is a sectional view illustrating a configuration example of a portion of the semiconductor package 1 according to the first embodiment. FIG. 2 illustrates a cross section of stacked two memory chips 40_1 and 40_2. The memory chip 40_1 and the memory chip 40_2 are bonded to each other on a bonding surface B_chip.

The memory chip 40_1 includes an array chip CH_A1 including a memory cell array MCA1, and a circuit chip CH_C1 including a CMOS (Complementary Metal Oxide Semiconductor) circuit CMOS1. The memory chip 40_2 includes an array chip CH_A2 including a memory cell array MCA2, and a circuit chip CH_C2 including a CMOS circuit CMOS2. In the memory chip 40_1, one of the memory cell array MCA1 and the CMOS circuit CMOS1 may be a first semiconductor circuit and the other thereof may be a second semiconductor circuit. In the memory chip 40_2, one of the memory cell array MCA2 and the CMOS circuit CMOS2 may be a first semiconductor circuit and the other thereof may be a second semiconductor circuit. The memory cell array MCA1 and the CMOS circuit CMOS1 included in the memory chip 40_1 may be a first semiconductor circuit and the memory cell array MCA2 and the CMOS circuit CMOS2 included in the memory chip 40_2 may be a second semiconductor circuit with the bonding surface B_chip interposed therebetween.

Memory Chip 40_1

The array chip CH_A1 includes the memory cell array MCA1 covered with an interlayer dielectric film ILD1_1. The memory cell array MCA1 includes a plurality of word lines WL1 stacked in a Z direction and insulated from each other, and a plurality of columnar bodies CL1 extending to penetrate through the stacked word lines WL1 in the stacking direction (the Z direction). Memory cells MC1 are provided to correspond to intersections between the word lines WL1 and the columnar bodies CL1, respectively. One ends of the columnar bodies CL1 are connected in common to a source line SL1. The other ends of the columnar bodies CL1 are connected to any of bit lines BL1 extending in a Y direction, respectively.

The memory cell array MCA1 is provided in an array region R_Arr. The word lines WL1 extend in an X direction to a terrace region R_Trr and are formed in the manner of stairs in the terrace region R_Trr. A contact plug CC1 is connected to a step surface of each of the word lines WL1 formed in the stair manner. Each of the contact plugs CC1 is electrically connected between an associated one of pads P1_1a provided in the terrace region R_Trr and the associated word line WL1 via a wiring layer W1_1. The pads P1_1a are electrode pads exposed on a surface of the interlayer dielectric film ILD1_1 and provided on a bonding surface B_mc1 of the array chip CH_A1. The wiring layer W1_1 electrically connects between the memory cell array MCA1 and the pads P1_1a via the contact plugs CC1, respectively.

A peripheral region R_Pri is provided around the array region R_Arr and the terrace region R_Trr. The peripheral region R_Pri may be provided at various locations including a central portion of the memory chips as well as the peripheral portion of the memory chips. Contact plugs Cpri1 are provided in the peripheral region R_Pri to penetrate through the interlayer dielectric film ILD1_1 of the array chip CH_A1 in the Z direction. One ends of the contact plugs Cpri1 are electrically connected to the pads P1_1a provided on the bonding surface B_mc1 in the peripheral

region R_Pri via the wiring layer W1_1, respectively. The other ends of the contact plugs Cpri1 are electrically connected to pads P1_1b provided on the opposite surface to the bonding surface B_mc1 of the array chip CH_A1, respectively.

The circuit chip CH_C1 is provided below (in a -Z direction) the array chip CH_A1 and includes the CMOS circuit CMOS1 covered with an interlayer dielectric film ILD1_2. The CMOS circuit CMOS1 is a circuit provided on a semiconductor layer SUB1 and including a P-type MOSFET (MOS Field Effect Transistor) and an N-type MOSFET. The CMOS circuit CMOS1 may include other semiconductor elements (for example, a resistive element or a capacitive element). The CMOS circuit CMOS1 is covered with the interlayer dielectric film ILD1_2. A multilayer wiring layer W1_2 is provided in the interlayer dielectric film ILD1_2. The multilayer wiring layer W1_2 electrically connects between the CMOS circuit CMOS1 and pads P1_2a. The pads P1_2a are electrode pads exposed on a surface of the interlayer dielectric film ILD1_2 and provided on the bonding surface B_mc1 of the circuit chip CH_C1. The pads P1_2a may be provided in any of the array region R_Arr, the terrace region R_Trr, and the peripheral region R_Pri.

Through silicon vias TSV1 are provided in the peripheral region R_Pri of the circuit chip CH_C1. The through silicon vias TSV1 are portions of the electrodes 50, respectively. The through silicon vias TSV1 penetrate through the semiconductor layer SUB1 in the Z direction and are electrically connected between associated ones of the pads P1_2a and associated ones of pads P1_2b, respectively. The pads P1_2b are electrode pads provided at end portions of the through silicon vias TSV1 on the opposite side to the bonding surface B_mc1, respectively.

The array chip CH_A1 and the circuit chip CH_C1 are stuck to each other on the bonding surface B_mc1. The interlayer dielectric films ILD1_1 and ILD1_2 are bonded and the pads P1_1a and P1_2a are bonded, respectively, on the bonding surface B_mc1. This enables the CMOS circuit CMOS1 of the circuit chip CH_C1 to be electrically connected to the memory cell array MCA1 via the multilayer wiring layer W1_2, the pads P1_2a and P1_1a, and the contact plugs CC1. As a result, the CMOS circuit CMOS1 can control the memory cell array MCA1. The through silicon vias TSV1 are electrically connected to the contact plugs Cpri1 via the pads P1_2a and P1_1a and the wiring layer W1_1. The through silicon vias TSV1 are provided, for example, to enable the supply power or the ground potential to be transmitted to chips in common.

Memory Chip 40_2

The array chip CH_A2 includes the memory cell array MCA2 covered with an interlayer dielectric film ILD2_1. The memory cell array MCA2 includes a plurality of word lines WL2 stacked in the Z direction and insulated from each other, and a plurality of columnar bodies CL2 extending to penetrate through the stacked word lines WL2 in the stacking direction (the Z direction). Memory cells MC2 are provided to correspond to intersections between the word lines WL2 and the columnar bodies CL2, respectively. One ends of the columnar bodies CL2 are connected in common to a source line SL2. The other ends of the columnar bodies CL2 are connected to any of bit lines BL2 extending in the Y direction, respectively.

The memory cell array MCA2 is provided in the array region R_Arr. The word lines WL2 extend in the X direction to the terrace region R_Trr and are formed in the manner of stairs in the terrace region R_Trr. A contact plug CC2 is connected to a step surface of each of the word lines WL2

formed in the stair manner. Each of the contact plugs CC2 is electrically connected between an associated one of pads P2_1a provided in the terrace region R_Trr and the associated word line WL2 via a wiring layer W2_1. The pads P2_1a are electrode pads exposed on a surface of the interlayer dielectric film ILD2_1 and provided on a bonding surface B_mc2 of the array chip CH_A2. The wiring layer W2_1 electrically connects between the memory cell array MCA2 and the pads P2_1a via the contact plugs CC2, respectively.

The peripheral region R_Pri is provided around the array region R_Arr and the terrace region R_Trr. Contact plugs Cpri2 are provided in the peripheral region R_Pri to penetrate through the interlayer dielectric film ILD2_1 of the array chip CH_A2 in the Z direction. One ends of the contact plugs Cpri2 are electrically connected to the pads P2_1a provided on the bonding surface B_mc2 in the peripheral region R_Pri via the wiring layer W2_1, respectively. The other ends of the contact plugs Cpri2 are electrically connected to pads P2_1b provided on the opposite surface to the bonding surface B_mc2 of the array chip CH_A2, respectively.

The circuit chip CH_C2 is provided below (in the -Z direction) the array chip CH_A2 and includes the CMOS circuit CMOS2 covered with an interlayer dielectric film ILD2_2. The CMOS circuit CMOS2 is a circuit provided on a semiconductor layer SUB2 and including a P-type MOSFET and an N-type MOSFET. The CMOS circuit CMOS2 may include other semiconductor elements (for example, a resistive element or a capacitive element). The CMOS circuit CMOS2 is covered with the interlayer dielectric film ILD2_2. A multilayer wiring layer W2_2 is provided in the interlayer dielectric film ILD2_2. The multilayer wiring layer W2_2 electrically connects between the CMOS circuit CMOS2 and pads P2_2a. The pads P2_2a are electrode pads exposed on a surface of the interlayer dielectric film ILD2_2 and provided on the bonding surface B_mc2 of the circuit chip CH_C2. The pads P2_2a may be provided in any of the array region R_Arr, the terrace region R_Trr, and the peripheral region R_Pri.

Through silicon vias TSV2 are provided in the peripheral region R_Pri of the circuit chip CH_C2. The through silicon vias TSV2 penetrate through the semiconductor layer SUB2 in the Z direction and are electrically connected between associated ones of the pads P2_2a and associated one of pads P2_2b, respectively. The pads P2_2b are electrode pads provided at end portions of the through silicon vias TSV2 on the opposite side to the bonding surface B_mc2, respectively.

The array chip CH_A2 and the circuit chip CH_C2 are stuck to each other on the bonding surface B_mc2. The interlayer dielectric films ILD2_1 and ILD2_2 are bonded and the pads P2_1a and P2_2a are bonded, respectively, on the bonding surface B_mc2. This enables the CMOS circuit CMOS2 of the circuit chip CH_C2 to be electrically connected to the memory cell array MCA2 via the multilayer wiring layer W2_2, the pads P2_2a and P2_1a, and the contact plugs CC2. As a result, the CMOS circuit CMOS2 can control the memory cell array MCA2. The through silicon vias TSV2 are electrically connected to the contact plugs Cpri2 via the pads P2_2a and P2_1a and the wiring layer W2_1. The through silicon vias TSV2 are also provided, for example, to enable the supply power or the ground potential to be transmitted to chips in common.

Bonding Between Memory Chips 40_1 And 40_2

The memory chip 40_1 and the memory chip 40_2 are bonded to each other on the bonding surface B_chip. The

pads **P1_2b** and the pads **P2_1b** are respectively bonded to each other on the bonding surface **B_chip**. The memory chips **40_1** and **40_2** are electrically connected via the pads **P1_2b** and **P2_1b** bonded to each other. Accordingly, the through silicon vias **TSV1** and **TSV2** and the contact plugs **Cpri1** and **Cpri2** are electrically connected and can transmit, for example, the supply power or the ground potential to the stacked memory chips **40_1** and **40_2** in common.

Configuration Of Pads **P1_1a** And The Like

FIG. 3A is a plan view illustrating a configuration example of the pads **P1_1a**. In FIG. 3A, the pads **P1_1a** are exposed from the surface of the interlayer dielectric film **ILD1_1** in a first planar view from a direction substantially perpendicular to the surface (the bonding surface **B_mc1**) of the interlayer dielectric film **ILD1_1** of the array chip **CH_A1** (a planar view seen from the Z direction). In the above planar view, each of the pads **P1_1a** is surrounded by the interlayer dielectric film **ILD1_1** and has, for example, a substantially octagonal shape. The planar shape of each of the pads **P1_1a** may be a shape of a polygon other than the octagon, a substantially circular shape, or a substantially elliptical shape.

A barrier metal film **101_1a**, a conductive material **102_1a**, and an insulating material **103_1a** are provided on the inner side of each of the pads **P1_1a**.

The barrier metal film **101_1a** is provided at the outer edge of each of the pads **P1_1a** and is located between the interlayer dielectric film **ILD1_1** or the insulating material **103_1a** and the conductive material **102_1a**. For example, a conductive material such as a stacked film including a titanium film and a titanium nitride film is used as the barrier metal film **101_1a**.

The conductive material **102_1a** is provided on the inner side of each of the pads **P1_1a** surrounded by the barrier metal film **101_1a**. For example, a conductive material such as copper or tungsten is used as the conductive material **102_1a**. The insulating material **103_1a** is provided in the manner of islands on the inner side of the conductive material **102_1a** and is surrounded by the conductive material **102_1a**.

In the above planar view, a plurality of the insulating materials **103_1a** extend in the Y direction and each have an elongated shape on a surface of the conductive material **102_1a**. In the above planar view, the insulating materials **103_1a** are arrayed in the X direction orthogonal to the Y direction in the manner of stripes or the manner of lines and spaces on the surface of the conductive material **102_1a**. In other words, the insulating materials **103_1a** are provided in the manner of slits or strips extending substantially in parallel to each other. The insulating materials **103_1a** are provided on the inner side of each of the pads **P1_1a** and do not reach the barrier metal film **101_1a** and the interlayer dielectric film **ILD1_1** in the above planar view. The insulating materials **103_1a** may connect to the interlayer dielectric film **ILD1_1** below the pads **P1_1a**. The same material (for example, a silicon dioxide film) as that of the interlayer dielectric film **ILD1_1** can be used as the insulating materials **103_1a**.

In the above planar view, the area of the insulating materials **103_1a** in each of the pads **P1_1a** is smaller than the area of the conductive material **102_1a**. By setting the area of the conductive material **102_1a** to be relatively large, the contact area with the conductive material **102_2a** in the pads **P1_2a** of the circuit chip **CH_C1** becomes larger and the contact resistance between each of the pads **P1_1a** and the associated one of the pads **P1_2a** can be suppressed to be low.

The insulating materials **103_1a** are formed of a material lower in the etching rate in the CMP process than the material (for example, a metallic material such as copper or tungsten) of the conductive material **102_1a** (for example, an oxide film such as a silicon dioxide film, a nitride film such as a silicon nitride film, a carbide film such as a silicon carbide film, or a composite material thereof can be used as the insulating materials **103_1a**). For example, the insulating materials **103_1a** can be formed of a material physically harder and less likely to be polished than the material of the conductive material **102_1a**. Alternatively, the insulating materials **103_1a** may be formed of a material less likely to be chemically etched by an abrasive (slurry) than the material of the conductive material **102_1a**. Therefore, in the CMP process, the insulating materials **103_1a** serve as supporting posts on the inner side of the conductive material **102_1a** and can reduce thinning of the film thickness at a central portion of the conductive material **102_1a** and reduce production of a recess (dishing).

A width **Wp1_1a** in the X direction or the Y direction of each of the pads **P1_1a** is, for example, about 1 micrometer (μm). A width **W103_1a** of each of the insulating materials **103_1a** is, for example, about several tens of nanometers.

The wiring layer **W1_1** indicated by a broken line is provided below each of the pads **P1_1a**. The wiring layer **W1_1** is electrically connected to the associated pad **P1_1a** through via contacts **V1_1**. In the present embodiment, nine via contacts **V1_1** are provided between each of the pads **P1_1a** and the associated wiring layer **W1_1**. However, the number of the via contacts **V1_1** is not limited to nine and can be any value. FIG. 3C is a plan view illustrating one example of the configuration of the wiring layer **W1_1**. The wiring layer **W1_1** is formed in the shape of a cross within a substantially square frame below the associated pad **P1_1a** in the planar view described above. The nine via contacts **V1_1** are provided on the wiring layer **W1_1**. The wiring layer **W1_1** may have a solid shape instead of the cross shape.

FIG. 3B is a sectional view illustrating a configuration example of the pads **P1_1**. FIG. 3B illustrates a cross section along a line B-B in FIG. 3A. The pads **P1_1a** are embedded in the interlayer dielectric film **ILD1_1** and are exposed on the surface of the interlayer dielectric film **ILD1_1**. The conductive material **102_1a** is electrically connected to the associated wiring layer **W1_1** provided therebelow through the via contacts **V1_1**.

The insulating materials **103_1a** can be parts of the interlayer dielectric film **ILD1_1** and can be of the same material. The height of the conductive material **102_1a** is, for example, about 1 μm .

As described above, according to the present embodiment, each of the pads **P1_1a** includes the insulating materials **103_1a** provided in the manner of islands on the inner side of the conductive material **102_1a** in the planar view from the direction substantially perpendicular to the bonding surface **B_mc1**. The insulating materials **103_1a** are formed of a material lower in the etching rate than the conductive material **102_1a**.

Therefore, in the CMP process to polish the interlayer dielectric film **ILD1_1** and the conductive material **102_1a**, the insulating materials **103_1a** serve as supporting posts within the conductive material **102_1a** and can reduce dishing of the conductive material **102_1a**.

If the insulating materials **103_1a** are not provided, the conductive material **102_1a** is polished in a relatively wide area. In this case, the inner side of the conductive material **102_1a** is greatly dished and recessed.

In contrast thereto, according to the present embodiment, the insulating materials **103_1a** divide the conductive material **102_1a** into relatively small areas and serve as supporting posts within the conductive material **102_1a**. Accordingly, dishing is suppressed on the inner side of the conductive material **102_1a**.

It is preferable that the insulating materials **103_1a** are arranged substantially uniformly in the conductive material **102_1a**. This can suppress dishing of the conductive material **102_1a** from occurring locally greatly.

While the pads **P1_1a** have been explained with reference to FIGS. 3A and 3B, the pads **P1_2a**, **P2_1a**, **P2_2a**, **P1_2b**, and **P2_1b** can be similarly configured. Therefore, dishing is suppressed in the CMP process also in pads **P1_2a**, **P2_1a**, **P2_2a**, **P1_2b**, and **P2_1b** other than the pads **P1_1a**. Since the configurations of the pads **P1_2a**, **P2_1a**, **P2_2a**, **P1_2b**, and **P2_1b** can be easily understood with reference to FIGS. 3A and 3B, detailed explanations thereof are omitted.

FIG. 4 is a sectional view illustrating a configuration example of a portion of the bonding surface **B_mc1**. The pads **P1_1a** on the side of the array chip **CH_A1** and the pads **P1_2a** on the side of the circuit chip **CH_C1** are respectively bonded to each other on the bonding surface **B_mc1**.

The pads **P1_1a** and the pads **P1_2a** both have the configuration illustrated in FIGS. 3A and 3B. Therefore, similarly to the pads **P1_1a** illustrated in FIG. 3A, the barrier metal film **101_2a**, the conductive material **102_2a**, and the insulating material **103_2a** are provided on the inner side of each of the pads **P1_2a** in a planar view from a direction substantially perpendicular to the surface of the interlayer dielectric film **ILD1_2**.

The pads **P1_1a**, the interlayer dielectric film **ILD1_1**, the barrier metal film **101_1a**, the conductive material **102_1a**, and the insulating material **103_1a** in FIGS. 3A and 3B are respectively read as the pads **P1_2a**, the interlayer dielectric film **ILD1_2**, the barrier metal film **101_2a**, the conductive material **102_2a**, and the insulating material **103_2a** in the following explanations of the pads **P1_2a**.

The barrier metal film **101_2a** is provided at the outer edge of each of the pads **P1_2a** and is located between the interlayer dielectric film **ILD1_2** or the insulating material **103_2a** and the conductive material **102_2a**. For example, a conductive material such as a stacked film including a titanium film and a titanium nitride film is used as the barrier metal film **101_2a**.

The conductive material **102_2a** is provided on the inner side of each of the pads **P1_2a** surrounded by the barrier metal film **101_2a**. For example, a conductive material such as copper or tungsten is used as the conductive material **102_2a**. The insulating material **103_2a** is provided in the manner of islands on the inner side of the conductive material **102_2a** and is surrounded by the conductive material **102_2a**.

In the planar view described above, a plurality of the insulating materials **103_2a** extend in the Y direction and each have an elongated shape on a surface of the conductive material **102_2a**. In the above planar view, the insulating materials **103_2a** are arrayed in the X direction orthogonal to the Y direction in the manner of stripes or the manner of lines and spaces on the surface of the conductive material **102_2a**. In other words, the insulating materials **103_2a** are provided in the manner of slits or strips extending substantially in parallel to each other. The insulating materials **103_1a** are provided on the inner side of each of the pads **P1_2a** and do not reach the barrier metal film **101_2a** and the interlayer dielectric film **ILD1_2** in the above planar

view. The same material (for example, a silicon dioxide film) as that of the interlayer dielectric film **ILD1_2** can be used as the insulating materials **103_2a**.

The insulating materials **103_2a** are formed of a material (for example, a silicon dioxide film) lower in the etching rate than the material (for example, a metallic material such as copper or tungsten) of the conductive material **102_2a**. For example, the insulating materials **103_2a** may be formed of a material physically harder and less likely to be polished than the material of the conductive material **102_2a**. Alternatively, the insulating materials **103_2a** may be formed of a material less likely to be chemically etched by an abrasive (slurry) than the material of the conductive material **102_2a**. Therefore, in the CMP process, the insulating materials **103_2a** serve as supporting posts on the inner side of the conductive material **102_2a** and can reduce dishing of the conductive material **102_2a**.

A width **Wp1_2a** in the X direction or the Y direction of each of the pads **P1_2a** is, for example, about 1 μm . A width **W103_2a** of each of the insulating materials **103_2a** is, for example, about several tens of nanometers.

The pads **P1_2a** are embedded in the interlayer dielectric film **ILD1_2** and are exposed on the surface of the interlayer dielectric film **ILD1_2**. The conductive material **102_2a** is electrically connected to the associated wiring layer **W1_2** provided therebelow. The insulating materials **103_2a** can be parts of the interlayer dielectric film **ILD1_2** and can be of the same material. The height of the conductive material **102_2a** is, for example, about 1 μm .

As described above, the pads **P1_1a** and **P1_2a** have substantially same configurations. Each of the pads **P1_1a** and the associated pad **P1_2a** are bonded to each other on the bonding surface **B_mc1** between the pads **P1_1a** and the pads **P1_2a** in such a manner that the extending direction of the insulating materials **103_1a** and the extending direction of the insulating materials **103_2a** are substantially same directions (for example, the Y direction). Accordingly, the conductive material **102_1a** and the conductive material **102_2a** are bonded to substantially face each other and match each other on the bonding surface **B_mc1** as illustrated in FIG. 4 when the array chip **CH_A1** is stuck to the circuit chip **CH_C1**. At that time, the pads **P1_1a** and the pads **P1_2a** are little dished and the conductive materials **102_1a** and **102_2a** are little recessed on the bonding surface **B_mc1**. That is, the conductive material **102_1a** and **102_2a** are provided to be substantially flush on the bonding surface **B_mc1**. Therefore, the conductive material **102_1a** and the conductive material **102_2a** can be bonded on the bonding surface **B_mc1** with a sufficiently low resistance while the insulating materials **103_1a** and **103_2a** are provided on the inner side of the conductive material **102_1a** and the conductive material **102_2a**, respectively.

If the insulating materials **103_1** and **103_2** are not provided, the conductive materials **102_1a** and **102_2a** are likely to be poorly bonded due to dishing in the CMP process while the areas of the conductive materials **102_1a** and **102_2a** on the bonding surface **B_mc1** are correspondingly increased. Therefore, there is a risk that the contact resistance between the conductive material **102_1a** and the conductive material **102_2a** is increased.

In contrast thereto, according to the present embodiment, the insulating materials **103_1a** and **103_2a** are provided and therefore the areas of the conductive materials **102_1a** and **102_2a** on the bonding surface **B_mc1** are correspondingly decreased. However, dishing of the conductive materials **102_1a** and **102_2a** are suppressed and the conductive materials **102_1a** and **102_2a** are little recessed on the

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bonding surface B_mc1. Therefore, the contact resistance between the conductive material **102_1a** and the conductive material **102_2a** can be lowered and stabilized.

A manufacturing method of the pads **P1_1a** and **P1_2a** according to the first embodiment is explained next.

FIGS. 5 to 11 are sectional views illustrating one example of the manufacturing method of the pads **P1_1a** according to the first embodiment. Since the manufacturing method of the pads **P1_2a** is same as that of the pads **P1_1a**, detailed explanations thereof are omitted.

First, the memory cell array MCA1, the interlayer dielectric film **ILD1_1**, and the like are formed on a substrate (for example, a silicon substrate) for the array chip CH_A1. Next, the wiring layer **W1_1** is formed in the interlayer dielectric film **ILD1_1** of the array chip CH_A1.

Next, an insulating film is further deposited on the wiring layer **W1_1** and the interlayer dielectric film **ILD1_1**. The insulating film can be of the same material (for example, a silicon dioxide film) as that of the interlayer dielectric film **ILD1_1**. Therefore, the insulating film on the wiring layer **W1_1** is also referred to as the interlayer dielectric film **ILD1_1**. The structure illustrated in FIG. 5 is thereby obtained.

Next, the interlayer dielectric film **ILD1_1** on the wiring layer **W1_1** is processed using a lithography technique and an etching technique. Accordingly, the interlayer dielectric film **ILD1_1** on the wiring layer **W1_1** is processed into a pattern of the via contacts **V1_1** as illustrated in FIG. 6.

Subsequently, a barrier metal film **201_1a** and a conductive material **202_1a** are deposited on the interlayer dielectric film **ILD1_1** and the wiring layer **W1_1** as illustrated in FIG. 7. For example, a stacked film including a titanium film and a titanium nitride film is used as the barrier metal film **201_1a**. For example, a conductive material such as copper or tungsten is used as the conductive material **202_1a**.

Next, the barrier metal film **201_1a** and the conductive material **202_1a** are polished using the CMP method until the interlayer dielectric film **ILD1_1** is exposed. The via contacts **V1_1** each including the barrier metal film **201_1a** and the conductive material **202_1a** are thereby formed as illustrated in FIG. 8.

Next, an insulating film is further deposited on the via contacts **V1_1**. The insulating film can be of the same material (for example, a silicon dioxide film) as that of the interlayer dielectric film **ILD1_1**. Therefore, the insulating film on the via contacts **V1_1** is also referred to as the interlayer dielectric film **ILD1_1**. Subsequently, the interlayer dielectric film **ILD1_1** on the via contacts **V1_1** is processed into a pattern of the pads **P1_1a** using a lithography technique and an etching technique as illustrated in FIG. 9. A first concave portion **Con_1** may be formed on each of the pads **P1_1a**, a first insulating layer may be formed around the first concave portion **Con_1**, and a first insulating material may be formed on the inner side of the first concave portion **Con_1**.

Next, the barrier metal film **101_1a** and the conductive material **102_1a** are deposited on the interlayer dielectric film **ILD1_1** and the via contacts **V1_1** as illustrated in FIG. 10. For example, a stacked film including a titanium film and a titanium nitride film is used as the barrier metal film **101_1a**. For example, a conductive material such as copper or tungsten is used as the conductive material **102_1a**.

Subsequently, the barrier metal film **101_1a** and the conductive material **102_1a** are polished using the CMP method until the interlayer dielectric film **ILD1_1** is exposed. Accordingly, the pads **P1_1a** each including the barrier metal film **101_1a** and the conductive material

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102_1a are formed as illustrated in FIG. 11. The interlayer dielectric film **ILD1_1** exposed in the CMP process becomes the insulating materials **103_1a** described above.

The insulating materials **103_1a** are provided on the inner side of the conductive material **102_1a** in the manner of islands (for example, the manner of stripes or the manner of lines and spaces) as illustrated in FIG. 3A. The insulating materials **103_1a** function as supporting posts in the conductive material **102_1a** in the CMP process for the barrier metal film **101_1a** and the conductive material **102_1a**. Accordingly, dishing (a recess) of the conductive material **102_1a** in the pads **P1_1a** is suppressed.

The manufacturing method of the pads **P1_1a** of the array chip CH_A1 has been explained above. The pads **P1_2a** of the circuit chip CH_C1 are formed in the same manner as that of the pads **P1_1a** while being connected to the CMOS circuit CMOS1. Therefore, dishing (a recess) of the conductive material **102_2a** is suppressed also in the pads **P1_2a**.

Dishing of the pads **P1_1a** of the array chip CH_A1 and the pads **P1_2a** of the circuit chip CH_C1 is suppressed. Therefore, when the array chip CH_A1 is stuck to the circuit chip CH_C1, the pads **P1_1a** and the pads **P1_2a** are sufficiently bonded to each other with almost no space therebetween as illustrated in FIG. 4. As a result, an increase in the contact resistance of the pads between the array chip CH_A1 and the circuit chip CH_C1 can be suppressed to suppress an open circuit failure.

While the bonding between the array chip CH_A1 and the circuit chip CH_C1 has been explained above, the present embodiment can be applied also to bonding between the memory chips **40_1** and **40_2**.

Bonding Between Memory Chips 40_1 And 40_2

As illustrated in FIG. 2, the memory chip **40_1** and the memory chip **40_2** are bonded to each other on the bonding surface B_chip. The memory chip **40_1** and **40_2** have a same configuration.

Each of the pads **P1_2b** of the memory chip **40_1** and the associated one of the pads **P2_1b** of the memory chip **40_2** are electrically connected to each other on the bonding surface B_chip. Each of the pads **P1_2b** is electrically connected to the associated one of the through silicon vias TSV1 provided in the circuit chip CH_C1 of the memory chip **40_1** via a redistribution layer (not illustrated). Each of the pads **P2_1b** is electrically connected to the associated one of the contact plugs Cpr2 of the array chip CH_A2 of the memory chip **40_2**.

The pads **P1_2b** and **P2_1b** can have the same configuration as that of the pads **P1_1a** illustrated in FIGS. 3A and 3B. Accordingly, each of the pads **P1_2b** and the associated one of the pads **P2_1b** are bonded in the same manner as the pads **P1_1a** and the pads **P1_2a** illustrated in FIG. 4. Therefore, the effects of the present embodiment can be achieved also in the bonding between the memory chips **40_1** and **40_2**.

FIGS. 12 to 17 are sectional views illustrating one example of a formation process of a region of the through silicon vias TSV1 of the circuit chip CH_C1.

First, the CMOS circuit CMOS1 is formed on a substrate (for example, a silicon substrate) SUB1 using a semiconductor manufacturing process. As illustrated in FIG. 12, the CMOS circuit CMOS1 is electrically connected to the through silicon vias TSV1 via the pads **P1_2a** and the wiring layer **W1_2** (or receiving electrodes for the through silicon vias), respectively. Illustrations of the CMOS circuit CMOS1, the pads **P1_2a**, and the wiring layer **W1_2** are omitted in FIG. 13 and subsequent drawings.

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Next, holes are formed in the formation region of the through silicon vias TSV1 using a lithography technique and an etching technique. A spacer dielectric film SP1 is formed on the inner walls of the holes. Next, the material (for example, copper or tungsten) of the through silicon vias TSV1 is embedded inside the spacer dielectric film (for example, a silicon dioxide film) SP1 using a plating method or the like. Subsequently, the interlayer dielectric film ILD1_2 is deposited on the substrate SUB1. The structure illustrated in FIG. 12 is thereby obtained.

In this way, the through silicon vias TSV1 are formed after the CMOS circuit is formed. Therefore, the through silicon vias TSV1 are formed after high-temperature heat treatment of the CMOS circuit, which enables the material (for example, copper or tungsten) of the through silicon vias TSV1 to be formed using the plating method. Ends of the through silicon vias TSV1 on the side of the CMOS circuit can be electrically connected to the CMOS circuit or may be electrically connected to an external electrode.

Next, the circuit chip CH_C1 is stuck to the array chip CH_A1. At that time, the pads P1_1a and the pads P1_2a are respectively bonded to each other (see FIG. 2).

Next, the substrate SUB1 is turned upside down as illustrated in FIG. 13. Subsequently, the back side of the substrate SUB1 is etched to expose ends of the through silicon vias TSV1 and the spacer dielectric film SP1 as illustrated in FIG. 14.

Next, insulating films 91 and 92 are deposited on the substrate SUB1 and the through silicon vias TSV1 as illustrated in FIG. 15. The insulating film 91 is, for example, a silicon nitride film and the insulating film 92 is, for example, a silicon dioxide film.

Next, the insulating films 91 and 92 are polished using the CMP method until the through silicon vias TSV1 are exposed as illustrated in FIG. 16. The through silicon vias TSV1 are thereby formed in the substrate SUB1. The through silicon vias TSV1 penetrate through the substrate SUB1 in a state of being electrically insulated from the substrate SUB1 by the spacer dielectric film SP1.

Next, a redistribution layer RW1 is formed as illustrated in FIG. 17. Next, the pads P1_2b are formed on the redistribution layer RW1. The configuration and formation method of the pads P1_2b are as explained with reference to FIGS. 3A to 11.

Subsequently, the memory chips 40_1 and 40_2 are stuck to each other. The pads P1_2b and the associated pads P2_1b are thereby respectively stuck to each other as illustrated in FIG. 2.

In a case in which through silicon vias are provided in the array chip CH_A1, the through silicon vias in the array chip CH_A1 can also be formed by the same method as illustrated in FIGS. 12 to 17.

First Modification

FIGS. 18A to 18D are sectional views illustrating another example of the manufacturing method of the pads P1_1a. Since the manufacturing method of the pads P1_2a is same as that of the pads P1_1a, detailed explanations thereof are omitted.

After the structure illustrated in FIG. 5 is formed, the interlayer dielectric film ILD1_1 on the wiring layer W1_1 is processed using a lithography technique and an etching technique. Accordingly, the interlayer dielectric film ILD1_1 on the wiring layer W1_1 is processed into the pattern of the via contacts V1_1 as illustrated in FIG. 18A.

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Next, the interlayer dielectric film ILD1_1 is processed again using the lithography technique and the etching technique to process an upper portion of the interlayer dielectric film ILD1_1 into the pattern of the pads P1_1a as illustrated in FIG. 18B. Accordingly, the pattern of the pads P1_1a is formed in the upper portion of the interlayer dielectric film ILD1_1 and the pattern of the via contacts V1_1 is formed under the pattern of the pads P1_1a to be continuous therewith.

Subsequently, the barrier metal film 101_1a and the conductive material 102_1a are deposited on the interlayer dielectric film ILD1_1 and the wiring layer W1_1 as illustrated in FIG. 18C.

Next, the barrier metal film 101_1a and the conductive material 102_1a are polished using the CMP method until the interlayer dielectric film ILD1_1 is exposed. Accordingly, the via contacts V1_1 and the pads P1_1a each including the barrier metal film 101_1a and the conductive material 102_1a are simultaneously formed as illustrated in FIG. 18D.

In this CMP process, the insulating materials 103_1a serve as supporting posts on the inner side of the conductive material 102_1a and can reduce dishing of the conductive material 102_1a.

The via contacts V1_1 and the pads P1_1a are simultaneously formed in the first modification. Therefore, in the first modification, the pads P1_1a can be formed in fewer processes than those in the first embodiment. The rest of the manufacturing process in the first modification can be same as that in the first embodiment. Accordingly, the first modification can achieve the effects identical to those of the first embodiment.

Second Modification

FIGS. 19A to 19G are sectional views illustrating still another example of the manufacturing method of the pads P1_1a. Since the manufacturing method of the pads P1_2a is same as that of the pads P1_1a, detailed explanations thereof are omitted.

After the structure illustrated in FIG. 5 is formed, the interlayer dielectric film ILD1_1 in the entire formation region of each of the pads P1_1a in the interlayer dielectric film ILD1_1 on the wiring layer W1_1 is removed using a lithography technique and an etching technique. The structure illustrated in FIG. 19A is thereby obtained.

Next, the barrier metal film 101_1a and the conductive material 102_1a are deposited on the interlayer dielectric film ILD1_1 and the wiring layer W1_1 as illustrated in FIG. 19B.

Next, the barrier metal film 101_1a and the conductive material 102_1a are polished using the CMP method until the interlayer dielectric film ILD1_1 is exposed. The barrier metal film 101_1a and the conductive material 102_1a are thereby formed in the entire formation region of each of the pads P1_1a as illustrated in FIG. 19C.

Subsequently, an upper portion of the conductive material 102_1a is processed using a lithography technique and an etching technique to remove the conductive material 102_1a located in the formation region of the insulating materials 103_1a. The structure illustrated in FIG. 19D is thereby obtained.

Next, a barrier metal film 101_3 is deposited on the interlayer dielectric film ILD1_1 and the conductive material 102_1a as illustrated in FIG. 19E.

Next, the insulating material 103_1a is deposited on the barrier metal film 101_3 as illustrated in FIG. 19F.

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Next, the insulating material **103_1a** is polished using the CMP method until the interlayer dielectric film **ILD1_1** is exposed. The pads **P1_1a** are thereby formed as illustrated in FIG. 19G. Even when the conductive material **102_1a** is exposed in this CMP process, the insulating materials **103_1a** serve as supporting posts on the inner side of the conductive material **102_1a** and dishing of the conductive material **102_1a** can be reduced.

In the second modification, the via contacts **V1_1** are formed in the entire formation region of each of the pads **P1_1a**. In this case, the pads **P1_1a** are connected to the wiring layer **W1_1** through the via contacts **V1_1**.

Also in the second modification, the via contacts **V1_1** and the pads **P1_1a** are simultaneously formed. Therefore, the pads **P1_1a** can be formed in fewer processes in the second modification than in the first embodiment. The rest of the formation process of the second modification can be same as that in the first embodiment. Accordingly, the second modification can achieve the effects identical to those of the first embodiment. An embodiment of using the pads **P1_1a** formed in the second modification will be described later with reference to FIG. 25.

Third Modification

FIGS. 20A to 20F are sectional views illustrating another example of the formation process of the region of the through silicon vias **TSV1** of the circuit chip **CH_C1**. In this modification, the through silicon vias **TSV1** are formed after the CMOS circuit is formed and the substrate **SUB1** is inverted.

First, the CMOS circuit (not illustrated) is formed on the substrate **SUB1** and the interlayer dielectric film **ILD1_2** is deposited thereon. The structure illustrated in FIG. 20A is thereby obtained.

Next, holes are formed in the formation region of the through silicon vias **TSV1** using a lithography technique and an etching technique as illustrated in FIG. 20B.

Subsequently, as illustrated in FIG. 20C, the spacer dielectric film **SP1** is formed on the inner wall of the holes and is etched back to remove the spacer dielectric film **SP1** on bottom portions of the holes.

Next, the material of the through silicon vias **TSV1** is embedded on the inner side of the spacer dielectric film **SP1** using a plating method or the like as illustrated in FIG. 20D.

In this way, the through silicon vias **TSV1** are formed after formation of the CMOS circuit. Accordingly, the through silicon vias **TSV1** are formed after high-temperature heat treatment of the CMOS circuit and therefore the material (for example, copper or tungsten) of the through silicon vias **TSV1** can be formed into a film using the plating method.

Next, the material of the through silicon vias **TSV1** is polished using the CMP method until the surface of the spacer dielectric film **SP1** is exposed. The through silicon vias **TSV1** are thereby formed in the substrate **SUB1** as illustrated in FIG. 20E. The through silicon vias **TSV1** penetrate through the substrate **SUB1** in a state of being electrically insulated from the substrate **SUB1** by the spacer dielectric film **SP1**.

Next, the redistribution layer **RW1** is formed as illustrated in FIG. 20F. Subsequently, the pads **P1_2b** are formed on the redistribution layer **RW1**. The configuration and formation method of the pads **P1_2b** are as explained with reference to FIGS. 3A to 11.

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Next, the memory chips **40_1** and **40_2** are stuck to each other. Accordingly, the pads **P1_2b** and the associated pads **P2_1b** are respectively stuck to each other as illustrated in FIG. 2.

In a case in which through silicon vias are provided in the array chip **CH_A1**, the through silicon vias in the array chip **CH_A1** can also be formed in the same manner as that in the present modification.

Second Embodiment

FIG. 21 is a plan view illustrating a configuration example of the pads **P1_2a** according to a second embodiment. In a case in which the pads **P1_1a** and the pads **P1_2a** have the same configuration as in the first embodiment, there is a possibility that the conductive material **102_1a** faces the insulating materials **103_2a** and the conductive material **102_2a** faces the insulating materials **103_1a** if the pads **P1_1a** are relatively displaced from the associated pads **P1_2a** in the X direction in FIG. 4, respectively. In this case, there is a risk that the contact area between the conductive material **102_1a** and the conductive material **102_2a** becomes extremely small, which increases and destabilizes the contact resistance between the pads **P1_1a** and the pads **P1_2a**.

In contrast thereto, the insulating materials **103_2a** of the pads **P1_2a** extend in a direction oblique to the X and Y directions in a planar view seen from the Z direction in the second embodiment. The configurations of the pads **P1_1a** may be identical to those of the first embodiment.

FIG. 22 is a sectional view illustrating a configuration example of a region of the bonding surface **B_mc1** according to the second embodiment. In the second embodiment, the pads **P1_1a** and the pads **P1_2a** are respectively bonded to each other in such a manner that the extending direction (for example, the Y direction) of the insulating materials **103_1a** intersects with the extending direction (a direction oblique to the X and Y directions) of the insulating materials **103_2a** on the bonding surface **B_mc1** when the array chip **CH_A1** and the circuit chip **CH_C1** are stuck to each other. A cross section along a line B-B in FIG. 21 is illustrated as the pad **P1_2a** in FIG. 22.

As viewed from a direction perpendicular to the substrate, the insulating materials **103_1a** separate from each other partially overlap with the conductive material **102_2a**, and the insulating materials **103_2a** separate from each other partially overlap with the conductive material **102_1a**. Since the extending direction of the insulating materials **103_1a** intersects with the extending direction of the insulating materials **103_2a**, the contact area between the conductive material **102_1a** and the conductive material **102_2a** does not reduce so much even if the pads **P1_1a** are displaced from the pads **P1_2a** to some extent in the X or Y direction. Therefore, the second embodiment enables the contact resistance to be low and stable against the displacement between the pads **P1_1a** and the pads **P1_2a** on the bonding surface **B_mc1**.

Third Embodiment

FIG. 23 is a plan view illustrating a configuration example of the pads **P1_1a** according to a third embodiment. In the third embodiment, the conductive material **102_1a** of each of the pads **P1_1a** has a mesh structure including elongated shapes extending in the X direction and elongated shapes extending in the Y direction on the surface of the interlayer dielectric film **ILD1_1** in a planar view seen from the Z

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direction. Therefore, the insulating materials **103_1a** are formed in the manner of islands (the manner of dots) and are arrayed two-dimensionally in a matrix in the X direction and the Y direction on the surface of the interlayer dielectric film **ILD1_1** in the planar view seen from the Z direction.

Assuming one of the insulating materials **103_1a** as a first insulating portion **In1_1**, an insulating portion arranged in the Y direction and closest to the first insulating portion **In1_1** is a second insulating portion **In2_1** and an insulating portion arranged in the X direction and closest to the first insulating portion **In1_1** is a third insulating portion **In3_1**.

The insulating materials **103_1a** are formed of a material (for example, a silicon dioxide film) lower in the etching rate than the material (for example, a metallic material such as copper or tungsten) of the conductive material **102_1a**. For example, the insulating materials **103_1a** can be formed of a material physically harder and less likely to be polished than the material of the conductive material **102_1a**. Alternatively, the insulating materials **103_1a** may be formed of a material less likely to be chemically etched by an abrasive (slurry) than the material of the conductive material **102_1a**. Therefore, in the CMP process, the insulating materials **103_1a** serve as supporting posts on the inner side of the conductive material **102_1a** and can reduce dishing of the conductive material **102_1a**.

The pads **P1_2a** also have the same configuration as that of the pads **P1_1a** in FIG. 23. Accordingly, the conductive material **102_2a** of each of the pads **P1_2a** also has a mesh structure including elongated shapes extending in the X direction and elongated shapes extending in the Y direction on the surface of the interlayer dielectric film **ILD1_2** in a planar view seen from the Z direction although not illustrated. That is, in the planar view seen from the Z direction, the insulating materials **103_2a** are formed in the manner of islands (the manner of dots) on the surface of the interlayer dielectric film **ILD1_2** and are arrayed two-dimensionally in a matrix in the X direction and the Y direction. Therefore, the insulating materials **103_2a** serve as supporting posts on the inner side of the conductive material **102_2a** in the CMP process and can reduce dishing of the conductive material **102_2a**.

This enables the contact resistance between the conductive material **102_1a** and the conductive material **102_2a** to be low and stable.

The third embodiment may be combined with any one of the first embodiment, the second embodiment, the first modification, and the second modification. That is, the pads **P1_1a** according to the third embodiment may be bonded to the pads **P1_2a** according to any one of the first embodiment, the second embodiment, the first modification, and the second modification.

The third embodiment may be used for bonding between the memory chips **40_1** and **40_2**. That is, the third embodiment may be applied to the pads **P1_2b** of the memory chip **40_1** and the pads **P2_1b** of the memory chip **40_2**. Accordingly, dishing of the pads **P1_2b** and the pads **P2_1b** is suppressed and the bonding between the memory chips **40_1** and **40_2** can also be stabilized with a low resistance.

Fourth Embodiment

FIG. 24 is a plan view illustrating a configuration example of the pads **P1_2a** according to a fourth embodiment. In the fourth embodiment, the conductive material **102_2a** of each of the pads **P1_2a** extends in a direction oblique to the X and Y directions in a planar view seen from the Z direction. The configurations of the pads **P1_1a** may be identical to those

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of any one of the first to third embodiments and the first and second modifications. The distance between dot-like insulating materials closest in the X direction and the Y direction in each of the pads **P1_1a** in the third embodiment is different from the distance between dot-like insulating materials closest in the X direction and Y direction when the conductive material **102_2a** is oblique as in the fourth embodiment. For example, assuming one of the insulating materials **103_2a** as a fourth insulating portion **In4_2**, an insulating portion arranged in the Y direction and closest to the fourth insulating portion **In4_2** is a fifth insulating portion **In5_2** and an insulating portion arranged in the X direction and closest to the fourth insulating portion **In4_2** is a sixth insulating portion **In6_2**. The distance between the first insulating portion **In1_1** and the second insulating portion **In2_1** as seen in the Y direction is shorter than the distance between the fourth insulating portion **In4_2** and the fifth insulating portion **In5_2**. The distance between the first insulating portion **In1_1** and the third insulating portion **In3_1** as seen in the X direction is shorter than the distance between the fourth insulating portion **In4_2** and the sixth insulating portion **In6_2**.

In the fourth embodiment, the pads **P1_1a** and the associated pads **P1_2a** are respectively bonded to each other in such a manner that the extending direction of the conductive material **102_1a** intersects with the extending direction of the conductive material **102_2a** on the bonding surface **B_mc1** when the array chip **CH_A1** and the circuit chip **CH_C1** are stuck to each other. Since the extending direction of the insulating materials **103_1a** intersects with the extending direction of the insulating materials **103_2a**, the contact area between the conductive material **102_1a** and the conductive material **102_2a** does not change so much even if the pads **P1_1a** are displaced to some extent in the X or Y direction from the pads **P1_2a**. Therefore, the fourth embodiment can stabilize the contact resistance against the displacement of the pads **P1_1a** from the pads **P1_2a** on the bonding surface **B_mc1**.

Other configurations of the fourth embodiment may be identical to the corresponding ones of the first to third embodiments. Accordingly, the fourth embodiment can also achieve the effects of any one of the first to third embodiments.

Fifth Embodiment

FIG. 25 is a sectional view illustrating a configuration example of the region of the bonding surface **B_mc1** according to a fifth embodiment. The pads **P1_1a** and **P1_2a** formed according to the second modification described above are used in the fifth embodiment.

In the fifth embodiment, the via contacts **V1_1** are provided below each of the pads **P1_1a** and are electrically connected in common to the conductive material **102_1a**. The via contacts **V1_1** electrically connect the conductive material **102_1a** to the wiring layer **W1_1**. In this way, the via contacts **V1_1** are provided to be integral with the conductive material **102_1a** in the entire formation region of each of the pads **P1_1a**.

Accordingly, the conductive material **102_1a** of the pads **P1_1a** is slightly raised from the bonding surface **B_mc1** due to volume expansion (heat expansion) of the via contacts **V1_1** and the conductive material **102_1a**.

Also for the pads **P1_2a**, the via contacts **V1_2** are similarly provided below each of the pads **P1_2a** and are electrically connected in common to the conductive material **102_2a**. The via contacts **V1_2** electrically connect the

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conductive material **102_2a** to the wiring layer **W1_2**. In this way, the via contacts **V1_2** are also provided in the entire formation region of each of the pads **P1_2a** to be integral with the conductive material **102_2a**. Accordingly, the conductive material **102_2a** of each of the pads **P1_2a** is slightly raised from the bonding surface **B_mc1** due to volume expansion (heat expansion) of the via contacts **V1_2** and the conductive material **102_2a**.

With raising of the pads **P1_1a** and **P1_2a** from the bonding surface **B_mc1**, the pads **P1_1a** and **P1_2a** are respectively reliably bonded to each other on the bonding surface **B_mc1**. Therefore, the pads **P1_1a** and **P1_2a** can be connected to each other stably with a low resistance.

While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

The invention claimed is:

1. A semiconductor device comprising:

a substrate having a first semiconductor circuit provided thereon;

first pads located on the substrate;

second pads respectively bonded to the first pads;

a first insulating layer located outside of each of the first pads on a bonding surface of the first pads and the second pads;

and

a second insulating layer located outside of each of the second pads on the bonding surface and bonded to the first insulating layer, wherein

the first pads each comprise a first area including a first conductor and a second area including a third insulating layer on the bonding surface, the second area is located inside of the first area on the bonding surface,

the first semiconductor circuit comprises one of a memory cell array and a CMOS circuit configured to control the memory cell array, and

the device further comprises a second semiconductor circuit comprising the other of the memory cell array and the CMOS circuit configured to control the memory cell array on an opposite side to the first semiconductor circuit with the bonding surface interposed between the first semiconductor circuit and the second semiconductor circuit.

2. The device of claim 1, wherein the second area comprises a first extending portion extending in a first direction on the bonding surface.

3. The device of claim 1, wherein the second area is smaller than the first area on the bonding surface.

4. The device of claim 1, wherein the second pads each comprise a third area including a second conductor and a fourth area including a fourth insulating layer on the bonding surface, the fourth area is located inside of the third area on the bonding surface.

5. The device of claim 4, wherein the fourth area comprises a second extending portion extending in a second direction lying in an in-plane direction of the bonding surface and different from the first direction on the bonding surface.

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6. The device of claim 1, wherein

on the bonding surface,

the third insulating layer comprises

a first insulating portion,

a second insulating portion arranged alongside of the first insulating portion in the first direction and being closest thereto, and

a third insulating portion arranged alongside of the first insulating portion in a second direction lying in an in-plane direction of the bonding surface and perpendicular to the first direction, and being closest thereto.

7. The device of claim 6, wherein

on the bonding surface,

the fourth insulating layer comprises

a fourth insulating portion,

a fifth insulating portion arranged alongside of the fourth insulating portion in the first direction and closest thereto, and

a sixth insulating portion arranged alongside of the fourth insulating portion in the second direction and closest thereto, and

when a distance between the first insulating portion and the second insulating portion closest in the first direction is a first distance, and

a distance between the fourth insulating portion and the fifth insulating portion closest in the first direction is a second distance,

the first distance is different from the second distance.

8. The device of claim 1, wherein the first insulating layer and the third insulating layer are simultaneously formed.

9. The device of claim 1, wherein

the first insulating layer and the first insulator comprise oxygen and silicon, and

the third insulating layer comprises copper, gold, or copper and gold.

10. A semiconductor device comprising:

a substrate having a first semiconductor circuit provided thereon;

first pads located on the substrate;

second pads respectively bonded to the first pads;

a first insulating layer located outside of each of the first pads on a bonding surface of the first pads and the second pads; and

a second insulating layer located outside of each of the second pads on the bonding surface and bonded to the first insulating layer, wherein

the first pads each comprise a first area including a first conductor and a second area including a third insulating layer on the bonding surface, the second area is located inside of the first area on the bonding surface,

the first semiconductor circuit comprises a first memory cell array and a first CMOS circuit configured to control the first memory cell array, and

the device further comprises a second semiconductor circuit comprising a second memory cell array and a second CMOS circuit configured to control the second memory cell array on an opposite side to the first semiconductor circuit with the bonding surface interposed between the first semiconductor circuit and the second semiconductor circuit.

11. The device of claim 10, wherein the second area comprises a first extending portion extending in a first direction on the bonding surface.

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12. The device of claim 10, wherein the second area is smaller than the first area on the bonding surface.

13. The device of claim 10, wherein the second pads each comprise a third area including a second conductor and a fourth area including a fourth insulating layer on the bonding surface, the fourth area is located inside of the third area on the bonding surface.

14. The device of claim 13, wherein the fourth area comprises a second extending portion extending in a second direction lying in an in-plane direction of the bonding surface and different from the first direction on the bonding surface.

15. The device of claim 10, wherein

on the bonding surface,

the third insulating layer comprises

a first insulating portion,

a second insulating portion arranged alongside of the first insulating portion in the first direction and being closest thereto, and

a third insulating portion arranged alongside of the first insulating portion in a second direction lying in an in-plane direction of the bonding surface and perpendicular to the first direction, and being closest thereto.

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16. The device of claim 15, wherein on the bonding surface,

the fourth insulating layer comprises a fourth insulating portion,

a fifth insulating portion arranged alongside of the fourth insulating portion in the first direction and closest thereto, and

a sixth insulating portion arranged alongside of the fourth insulating portion in the second direction and closest thereto, and

when a distance between the first insulating portion and the second insulating portion closest in the first direction is a first distance, and

a distance between the fourth insulating portion and the fifth insulating portion closest in the first direction is a second distance,

the first distance is different from the second distance.

17. The device of claim 10, wherein the first insulating layer and the third insulating layer are simultaneously formed.

18. The device of claim 10, wherein

the first insulating layer and the third insulating layer comprise oxygen and silicon, and

the first conductor comprises copper, gold, or copper and gold.

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